AP CSP Unit 6 Day 6.1 Binary, ASCII, Unicode Assessment

1. (DOK 1) How many bits are in one byte?

A. 4

B. 8

C. 16

D. 32

2. (DOK 1) Which number system uses only 0s and 1s?

A. Decimal

B. Hexadecimal

C. Binary

D. ASCII

3. (DOK 1) What does ASCII provide?

A. An image format

B. A character encoding standard

C. A programming language

D. A compression method

4. (DOK 1) What is the place value of the binary digit 1 in the rightmost position?

A. 1

B. 2

C. 4

D. 8

5. (DOK 2) What is the decimal value of binary 1010?

A. 8

B. 10

C. 12

D. 14

6. (DOK 2) Why was Unicode developed?

A. To replace the Internet

B. To represent many more characters and languages than ASCII

C. To store videos

D. To compress files

7. (DOK 2) How does UTF-8 improve text representation?

A. It supports variable-length encoding for Unicode characters

B. It only stores numbers

C. It removes binary

D. It uses exactly 64 bits per character

8. (DOK 3) A document contains English, Arabic, and Chinese characters. Which encoding choice is most appropriate and why?

A. ASCII because it uses fewer bits

B. Unicode with UTF-8 because it supports many writing systems

C. Binary because all files are binary

D. Decimal because it is easier to read

9. (DOK 3) A message encoded in ASCII contains a symbol that displays incorrectly. What is the best explanation?

A. ASCII may not include the required character

B. The file is not binary

C. The decimal system failed

D. The character uses too many bytes for storage

10. (DOK 4) A company needs to store customer names from countries around the world, exchange data between systems, and minimize storage for English text. Which solution best satisfies these requirements?

A. Use ASCII only

B. Use Unicode with UTF-8 encoding

C. Store all text as images

D. Convert everything to decimal numbers